

Software Requirements Specification (SRS)

Council for Legal Education and Training (CLET)

**GSL - Student Support Services + DTI (Digital
Transformation Initiative)**

**Student App - Mobile-First Student Front Door
(System S028)**

Legislative Basis: Legal Education Act, 2026 (Act 1170); (WCAG 2.1 AA accessibility - statutory)

Resource for Development (RfD)

Version 1.0 - July 2026

SYSTEMS

No.	System ID	System Name	Phase
1	S028	Student App (Mobile-First Student Front Door)	FAST-TRACK · MAPPED · DTI v3.0

REVISION HISTORY

Date	Document Version	Document Revision History	Document Author/Reviser
July 2026	1.0	Initial SRS compiled from the CLET_S028_StudentApp_Requirements workbook v1.0 (FAST-TRACK · MAPPED · DTI v3.0).	RfD + DTI

APPROVALS

Date	Document Version	Approver Name and Title	Approver Signature

REFERENCES

Ref ID	Document Title	Author/Source	Version	Date	Purpose / Description
REF-001	Legal Education Act, 2026 (Act 1170)	Republic of Ghana			Statutory basis mandating WCAG 2.1 AA accessibility compliance for the GSL Student App and its welfare,

					fee, and academic service flows.
REF-002	Comprehensive Digital System Mapping	CLET	1.0	2026	Comprehensive mapping of all the 252 digital systems required to operationalise the Council for Legal Education and Training (CLET) under the Legal Education Act, 2026 (Act 1170).

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1. Introduction

1.1. Purpose

S028 is the mobile-first, student-facing front door for the entire GSL student experience: admissions status, timetables, fees, results, hostel management, library access, welfare requests, scholarship applications, and communications. It is the single authenticated entry point to all student-facing CLET services, integrating with S027 NLEMS, S029 LMS, S141 Payment System, S104 Library System, and S031 Welfare CMS via the S026 API Gateway. WCAG 2.1 AA accessibility compliance is a statutory requirement under Act 47 across all 14 app modules. The app is engineered for offline-first to remain usable under Ghana's degraded-network conditions. This document defines the functional and non-functional requirements for S028 across its 14 SA modules, its authentication model, and its release governance.

1.2. Scope

In scope:

- Fourteen student-facing app modules (SA.01–SA.14) covering admissions status, timetable, attendance, grades and transcript, certificates, exam registration, appeals, notifications, fees and payments, hostel management, library services, welfare and counselling requests, scholarship applications, and co-curricular / society activity.
- Biometric authentication (fingerprint / Face ID) as the primary daily login method, with PIN fallback and TOTP step-up for sensitive actions, via S001 IAM SSO.
- Offline-first architecture: key screens (timetable, results, transcript, notifications inbox, library barcode, welfare crisis contacts) remain accessible via a cached local data layer, syncing when connectivity is restored.
- Statutory WCAG 2.1 AA accessibility compliance across all 14 modules, with an enhanced standard on SA.12 Welfare, under Legal Education Act, 2026 (Act 1170).
- Anonymous crisis access to welfare services (SA.12) without requiring login, with a persistent crisis button reachable in ≤ 2 taps from any screen.
- Integration with S027 NLEMS, S029 LMS, S141 Payment System, S104 Library System, S031 Welfare CMS, S019/S036 Certification Systems, S063 NBES, S009 Tribunal, and S043 Society CMS, all routed through the S026 API Gateway.

- An S028 Admin Portal for GSL Student Support Administrator content configuration (notification content, scholarship windows, welfare routing rules, hostel allocation), with S003 audit logging of all configuration changes.
- App Store and Google Play release governance, including mandatory WCAG 2.1 AA audits before each release and CLET DG step-up MFA approval for statutory-impacting releases.

Out of scope:

- Lecturer-facing and admin-facing functions, which are served by separate portals.
- Detailed society management administration (handled in S043) - S028 links to S043 for student-facing discovery only.
- Exam paper delivery, which is handled by S013 Exam Infrastructure for computer-based testing.
- Medical records beyond the submission of a welfare request.

1.3. Definitions, Acronyms and Abbreviations

1. Key acronyms used throughout this document are summarised below; the complete glossary is provided in Appendix A.
2. SA - the module code prefix for the 14 Student App screens (e.g. SA.01–SA.14).
3. 47 - Legal Education Act 47, the statutory basis mandating WCAG 2.1 AA accessibility for S028.
4. WCAG 2.1 AA - Web Content Accessibility Guidelines, Level AA; a statutory compliance target for S028, not a best-practice guideline.
5. DTI - Digital Transformation Initiative, S028's delivery and infrastructure co-owner.
6. NBE - National Bar Examination, administered via S063 NBES.
7. TOTP - Time-based One-Time Password, required for sensitive in-app actions.
8. SSO - Single Sign-On, provided via S001 IAM across all integrated backend systems.
9. Secure Enclave - the on-device hardware security module storing biometric data; never transmitted off the device.

2. Overall Description

2.1. Product Perspective

S028 is a mobile-first application (iOS and Android) that acts as the single authenticated front door to the CLET student digital ecosystem. It comprises 14 student-facing modules (SA.01–SA.14), a biometric-first authentication layer via S001 IAM, an offline-first local data cache, and a web-based Admin Portal for content configuration. All backend integration - to S027 NLEMS, S029 LMS, S141 Payment System, S104 Library System, S031 Welfare CMS, and other systems - routes exclusively through the S026 API Gateway, which enforces schema validation, field-level access control, and audit logging. WCAG 2.1 AA accessibility is engineered into every module from the outset as a statutory requirement, and an offline-first data architecture ensures core services remain usable under Ghana's variable network conditions.

2.2. Product Functions - Overview

S028 provides: biometric SSO authentication with anonymous crisis-access carve-out; real-time admissions status tracking; offline-capable timetable, attendance check-in, grades, transcript, and certificates; exam registration gated on real-time NBE eligibility; appeals and grade review submission; fee schedule display and payment initiation; hostel and library self-service; confidential, end-to-end encrypted welfare and counselling access; eligibility-filtered scholarship applications; a categorised notification inbox with statutory-notification protection; and an Admin Portal with app-store release governance for GSL Student Support and DTI.

2.3. User Classes and Characteristics

The primary users are four actor roles, each with a specific, least-privilege access scope enforced via S001 IAM.

Role	Description
Student / Applicant	The primary user of S028. Applicants track admissions status (SA.01) before enrolment; enrolled students use the full suite - timetable, attendance, grades, certificates, exam registration, appeals, fees, hostel, library, welfare, scholarships, and society activity. Read/write limited to their own data across all integrated systems.
GSL Student Support Administrator	Configures S028 content under GSL's product ownership: notification content, scholarship windows, welfare routing rules, hostel allocation rules, and the SA.01 admissions status workflow. No access to individual student data via S028 - analytics are aggregate only.

DTI Developer	Responsible for S028 build, deployment, and maintenance: app releases, S026 API Gateway configuration, push notification infrastructure, biometric authentication integration, and performance monitoring. No access to student personal data - infrastructure access only.
CLET DG	Approves major S028 feature changes affecting statutory obligations (accessibility, welfare, fees) and reviews the annual accessibility audit report. Read access to aggregate usage analytics only - not individual student usage data. Step-up MFA required for major release approval.

2.4. Status Matrix

No	Action	Applicant Status (What applicant sees)	Interpretation of Status	CLET (Internal Staff) Status (What Staff Sees)	Manual State Change (Yes/No)
1	Applicant submits admission application	Application received	Status tracked in SA.01, awaiting review	APPLICATION_RECEIVED	No
2	Admission decision made in S027	Offer made / rejected	Offer or rejection displayed with next steps; rejection includes appeal rights	DECISION_MADE	No
3	Applicant accepts offer and enrolls	Enrolled	Full SA.01–SA.14 suite becomes available	ENROLLED	No

4	Student check-in queued while offline	Check-in pending sync	Attendance check-in queued locally, syncs to S029 on reconnect	QUEUED_OFF LINE	No
5	Student initiates fee payment	Payment pending confirmation	TOTP-confirmed initiation; 60-second timeout window applies	PAYMENT_PENDING	Yes
6	Student submits welfare self-referral (anonymous or authenticated)	Referral submitted	S031 accepts referral and returns reference number; case status becomes OPEN	OPEN	No
7	Welfare case resolved and closed in S031	Case closed	Only OPEN/CLOSED status shown to student - no case content	CLOSED	No

2.5. Assumptions and Dependencies

Assumptions:

Dependencies:

- S001 IAM - biometric/PIN authentication, SSO, TOTP step-up, and session management.
- S026 API Gateway - the single integration point for all S028 backend calls, enforcing schema validation, field-level access control, rate limiting, and audit logging.

- S027 NLEMS - admissions status, timetable cohort assignment, grades, transcript, NBE eligibility, and academic standing for scholarships.
- S029 LMS - timetable feed and attendance module.
- S141 Payment System - fee schedule, payment processing, and receipts.
- S104 Library System - borrowed items, due dates, reservations, and fines.
- S031 Welfare CMS - welfare self-referral, case status, and secure messaging.
- S019 / S036 Certification Systems - certificate issuance and verification links.
- S063 NBES - Bar exam registration and eligibility.
- S025 Communications System and the Push Notification Service - all SA.08 notification delivery.
- S003 Audit - logging of all Admin Portal configuration changes and app release events.
- Students' devices support platform biometric APIs (Apple LocalAuthentication, Android BiometricPrompt); a PIN fallback covers devices without biometric hardware.
- S029 LMS, S031 Welfare CMS, and other backend systems expose the data required by each SA module via the S026 API Gateway on the FAST-TRACK delivery timeline.
- DTI maintains the app on both the Apple App Store and Google Play, including the 18-month backward compatibility window.
- Independent accessibility assessors are available to conduct the WCAG 2.1 AA audit required before each release.

2.6. System Use Case Overview

Four actors interact with S028 across the student digital experience: the Student / Applicant authenticates via biometric SSO (or anonymously for welfare crisis access) and uses the 14 SA modules for admissions, academics, fees, hostel, library, welfare, scholarships, and society activity; the GSL Student Support Administrator configures notification content, scholarship windows, welfare routing, and hostel rules via the web-based Admin Portal without ever accessing individual student data; the DTI Developer manages app releases, the S026 API Gateway, and infrastructure without access to student personal data; and the CLET DG approves major statutory-impacting releases via step-up MFA and reviews the annual accessibility audit and aggregate usage analytics. All backend data flows through the S026 API Gateway, WCAG 2.1 AA accessibility is enforced on every screen, and an offline-first cache keeps timetable, results, notifications, library access, and welfare crisis contacts available without connectivity.

3. Software Requirement Specifications

MODULE: Authentication & Onboarding (47)

FUNCTIONAL REQUIREMENT ID: S028-FR-A

SRS-S028-A-01: Biometric Authentication with On-Device Storage

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
A-01	S028 must provide biometric authentication (fingerprint / Face ID) as the primary daily login method after initial setup. Biometric data is stored on-device only in the Secure Enclave and never transmitted to any CLET server. PIN fallback is available on all devices; TOTP via S001 IAM is required for fee payment initiation, appeal submission, and welfare self-referral.	Biometric data never leaves the device; TOTP required on the three listed sensitive actions.	MUST

User Story

As a student, I want to log in with my fingerprint or Face ID, so that daily access to my student services is fast without compromising the security of my biometric data.

Requirements

- S028 provides biometric authentication (fingerprint / Face ID on supported devices) as the primary daily login method after initial setup.
- Biometric data is stored on-device only, in the device's Secure Enclave, and is never transmitted to any CLET server.
- PIN fallback is available on all devices.
- TOTP via S001 IAM is required for fee payment initiation, appeal submission, and welfare self-referral, regardless of biometric login status.

Validation

- S028 must use platform biometric APIs (Apple LocalAuthentication, Android BiometricPrompt) rather than custom biometric implementations.

Workflow

- Student completes initial S001 IAM SSO setup → enables biometric login → daily app access authenticated via on-device biometric check → sensitive actions additionally require TOTP step-up.

Acceptance Criteria

- Given a student sets up biometric login, when they authenticate daily, then no biometric data is transmitted off the device.
- Given a student initiates a fee payment, appeal, or welfare self-referral, when they proceed, then TOTP step-up is required regardless of biometric session status.

SRS-S028-A-02: Fast, Accessible Onboarding

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
A-02	Onboarding for a newly admitted student must be completable in under 5 minutes from the S025 admission letter deep link to a fully configured app profile, accessible to screen readers, with no steps requiring physical paperwork.	5-minute onboarding is both a usability and a statutory accessibility requirement under 47.	MUST

User Story

As a newly admitted student, I want to set up my app profile quickly and entirely on my phone, so that I can start using student services without delay or needing to print anything.

Requirements

- Onboarding for a newly admitted student is completable in under 5 minutes, starting from the S025 admission letter deep link to a fully configured app profile.
- The onboarding flow is accessible to screen readers (WCAG 2.1 AA).
- No onboarding step may require actions that cannot be completed on a mobile device (no 'print and sign' steps).

Validation

- The 5-minute onboarding target and screen-reader accessibility are both statutory requirements under Act 47, not just usability goals.

Workflow

- Student receives S025 admission letter with deep link → taps link → completes S001 IAM identity verification → configures profile and notification preferences → onboarding complete, all within 5 minutes.

Acceptance Criteria

- Given a newly admitted student starts onboarding via the deep link, when they complete it, then the process takes under 5 minutes and requires no physical paperwork.
- Given a student using a screen reader starts onboarding, when they proceed through each step, then all steps are fully accessible.

SRS-S028-A-03: Single Sign-On Across Backend Systems

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
A-03	S028 must support SSO via S001 IAM: once authenticated, the student accesses all integrated backend systems without re-authenticating. Session tokens expire after 24 hours of inactivity; sensitive actions still require TOTP step-up regardless of session status.	SSO reduces friction, critical for adoption in a low-bandwidth environment.	MUST

User Story

As a student, I want to authenticate once and access all my student services without repeated logins, so that using the app is frictionless even on a slow connection.

Requirements

- S028 supports SSO via S001 IAM: once authenticated, the student accesses S027 NLEMS, S029 LMS, S141 Payment, S104 Library, and S031 Welfare without re-authenticating.
- Session tokens expire after 24 hours of inactivity.
- Sensitive actions require TOTP step-up regardless of session status.

Validation

- SSO reduces login friction, which is critical for adoption in a low-bandwidth environment where repeated logins are a barrier to use.

Workflow

- Student authenticates once via S001 IAM → session token issued and used across all integrated backend systems → token expires after 24 hours of inactivity, requiring re-authentication → sensitive actions always require TOTP step-up.

Acceptance Criteria

- Given a student is authenticated, when they navigate between SA modules backed by different systems, then no re-authentication is required.
- Given 24 hours of inactivity pass, when the student returns, then re-authentication is required.

SRS-S028-A-04: Anonymous Crisis Access to Welfare Services

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
A-04	S028 must support anonymous crisis access: a student in crisis can reach SA.12 Welfare crisis intake and the S031 crisis button without logging in. Anonymous welfare referral is accepted by S031 with a reference number returned to the student; identity may optionally be provided afterward.	Anonymous crisis access is a safeguarding design requirement - login must never be a barrier.	MUST

User Story

As a student in crisis, I want to reach welfare support without needing to log in, so that distress or lack of access to my credentials never prevents me from getting help.

Requirements

- S028 supports **unauthenticated user/student** crisis access: a student in crisis can reach SA.12 Welfare & Counselling crisis intake and the S031 crisis button without logging in.
- Anonymous welfare referral is accepted by S031 with a reference number returned to the student.
- Identity may optionally be provided after the referral is made.

Validation

- This is a safeguarding design requirement: login barriers must never prevent a student in distress from reaching welfare services.

Workflow

- Student in crisis opens the crisis button without authenticating → submits anonymous referral → S031 accepts and returns a reference number → student may optionally add identity information afterward.

Acceptance Criteria

- Given a student is not logged in, when they access the crisis button, then they can still submit a welfare referral and receive a reference number.
- Given an anonymous referral is submitted, when S031 processes it, then a reference number is returned without requiring student identity.

MODULE: Academic Services (SA.01–SA.07)

FUNCTIONAL REQUIREMENT ID: S028-FR-B

SRS-S028-B-01: Real-Time Admissions Status Tracking

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-01	SA.01 Admissions Status must display the applicant's status from S027 NLEMS in real time with clear transitions (Application Received → Under Review → Decision Pending → Offer Made → Offer Accepted → Admitted). Each transition triggers an SA.08 notification; rejected applications display the decision with appeal rights.	Status display includes the S027 case reference for the applicant's records.	MUST

User Story

As an applicant, I want to track my admission status in real time, so that I always know exactly where my application stands.

Requirements

- SA.01 displays the applicant's current application status from S027 NLEMS in real time, with clear status transitions: Application Received → Under Review → Decision Pending → Offer Made → Offer Accepted → Enrolled.
- Each status transition triggers an SA.08 Notification.

- Rejected applications display the decision along with appeal rights.

Validation

- The status display must include the S027 case reference for the applicant's own records.

Workflow

- Application status changes in S027 NLEMS → SA.01 reflects the new status in real time → SA.08 notification triggered → if rejected, appeal rights are displayed alongside the decision.

Acceptance Criteria

- Given an application status change in S027, when the applicant opens SA.01, then the current status is displayed.
- Given an application is rejected, when the decision is shown, then appeal rights are displayed alongside it.

SRS-S028-B-02: Weekly Timetable with Offline Access and Calendar Export

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-02	SA.02 My Schedule must display the student's weekly timetable from S029 LMS (session name, lecturer, room, date/time, type). Timetable updates within 1 hour of any S029 change, is viewable weekly/daily, and is exportable to the device calendar (iCal).	Timetable cached for 7-day offline access; iCal export needs no server connectivity after download.	MUST

User Story

As a student, I want to view my weekly timetable and add it to my phone's calendar, so that I always know my schedule even without connectivity.

Requirements

- SA.02 displays the student's weekly timetable from S029 LMS: session name, lecturer, room/venue, date and time, and session type.
- The timetable updates within 1 hour of any change in S029.
- The timetable is viewable in weekly and daily formats and is exportable to the device calendar (iCal).

Validation

- The timetable is cached for 7-day offline access; iCal export does not require CLET server connectivity after the initial download.

Workflow

- S029 timetable data changes → SA.02 syncs the update within 1 hour → student views weekly/daily format → student exports to device calendar via iCal.

Acceptance Criteria

- Given a timetable change occurs in S029, when SA.02 syncs, then the update is reflected within 1 hour.
- Given a student is offline, when they open SA.02, then the last cached timetable (up to 7 days old) is displayed.

SRS-S028-B-03: QR and Geo-Based Attendance Check-In

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-03	SA.03 Attendance Check-in must support QR code scan and geo-based check-in (within a configurable radius). Check-in events are queued offline and synced to S029 LMS on reconnect.	QR and geo methods are institution-policy selectable; offline sync deduplication enforced at S029.	MUST

User Story

As a student, I want to check in to class quickly by scanning a QR code or via my location, so that marking attendance is fast even if I'm briefly offline.

Requirements

- SA.03 supports two check-in methods: QR code scan (lecturer-displayed code) and geo-based check-in (device location verified against the session's registered room GPS coordinates within a configurable radius).
- Check-in events are queued offline and synced to S029 LMS on reconnect.

Validation

- QR and geo methods are institution-policy selectable; offline sync deduplication is enforced at the S029 layer.

Workflow

- Student scans QR code or triggers geo check-in → if offline, event is queued locally → on reconnect, queued check-ins sync to S029 LMS → S029 deduplicates any repeated sync attempts.

Acceptance Criteria

- Given a student checks in while offline, when connectivity is restored, then the check-in syncs to S029 without duplication.
- Given a student uses geo check-in, when their location is outside the configured radius, then check-in is not accepted.

SRS-S028-B-04: Grades, Transcript, and Grade Review Initiation

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-04	SA.04 Grades & Transcript must display all assessment results from S027 NLEMS 27.10 (module, assessment type, mark, grade, cumulative GPA). The official transcript (27.12) is downloadable as PDF. Students can initiate a grade review, which routes to SA.07 Appeals.	Transcript PDF and grade data are accessible offline after first download/cache.	MUST

User Story

As a student, I want to view my grades and download my transcript, and request a review if something looks wrong, so that I have full visibility and recourse over my academic record.

Requirements

- SA.04 displays all assessment results from S027 NLEMS 27.10: module name, assessment type, mark, grade, and cumulative GPA.
- The official academic transcript (S027 27.12) is downloadable as a PDF from SA.04.
- Students can initiate a grade review from SA.04, which routes the request to SA.07 Appeals.

Validation

- The transcript PDF is accessible offline after first download; grade data is cached for offline viewing.

Workflow

- Student opens SA.04 → views grades and GPA → downloads transcript PDF (cached for offline use) → if disputing a grade, initiates a review that routes to SA.07.

Acceptance Criteria

- Given a student downloads their transcript, when they go offline, then the PDF remains accessible.
- Given a student initiates a grade review from SA.04, when submitted, then the request appears in SA.07 Appeals.

SRS-S028-B-05: Appeals, Progression Reviews, and Tribunal Referrals

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-05	SA.07 Appeals & Grade Reviews must allow grade review requests. If LLB it will be at the accredited institution (dean) , if LPT and it is NBE then it goes to NBES(CLET) and if it is academic term/year exam that one is settled at the Dean/institution registrar level	Appeal submission requires TOTP step-up; the 3-screen limit is both a usability and accessibility requirement.	MUST

User Story

As a student, I want to submit grade review requests, progression appeals, or examination-related appeals through a simple online process so that my request is automatically routed to the appropriate academic authority based on my programme and the type of appeal.

Requirements

The system shall allow students to submit:

- Grade Review Requests.
- Progression Appeals.
- Examination Appeals.

The system shall automatically route submissions as follows:

LLB students – Grade reviews and academic examination appeals shall be routed to the student's accredited institution (Dean/Registrar).

LPT students (National Bar Examination) – National Bar Examination appeals shall be routed to the National Bar Examinations Committee (NBES/CLET).

- **LPT students (Academic Term/Year Examinations)** – Appeals relating to academic term or year examinations shall be routed to the student's Dean/Institution Registrar.
- Each submission type shall be completed in three (3) screens or fewer.
- Upon successful submission, the system shall generate a confirmation message containing a unique reference number.

Validation

- Submission of any appeal or grade review request shall require TOTP step-up authentication before final submission.
- The maximum of three (3) screens for the submission process is both a usability and accessibility requirement.
- Mandatory supporting information and attachments shall be validated before submission.

Workflow

Student accesses SA.07 → Selects request type (Grade Review, Progression Appeal, or Examination Appeal) → Completes the request within three screens or fewer → Confirms submission using TOTP step-up authentication → System automatically determines the appropriate recipient based on programme and appeal type → Submission is routed to the responsible office → Student receives a confirmation with a unique reference number.

Acceptance Criteria

- **Given** a student submits a Grade Review, Progression Appeal, or Examination Appeal, **when** the submission is completed, then the system shall require TOTP step-up authentication and issue a unique reference number.
- **Given** an LLB student submits a Grade Review or academic examination appeal, **when** the request is submitted, then it shall be routed to the student's accredited institution (Dean/Registrar).

- **Given** an LPT student submits a National Bar Examination appeal, **when** the request is submitted, then it shall be routed to the National Bar Examinations Secretariat (NBES/CLET).
- **Given** an LPT student submits an academic term or year examination appeal, **when** the request is submitted, then it shall be routed to the student's Dean/Institution Registrar.
- **Given** the submission flow is measured, when a student completes the process, then it shall require no more than three (3) screens.
- **Given** mandatory information or required supporting documents are missing, **when** the student attempts to submit, then the system shall prevent submission and display appropriate validation messages.

SRS-S028-B-06: Certificates Wallet with Verifiable Sharing

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-06	SA.05 Certificates Wallet should display all S019/S036 certificates (LPT Certificate, Qualifying Certificate, course completion). Each certificate supports view, download (PDF-A), and share (time-limited S019.5 verification link).	Share link expires in 90 days and is renewable from SA.05.	SHOULD

User Story

As a student, I want to store and share my certificates on my phone, so that I can provide verifiable proof of my qualifications to employers or other institutions.

Requirements

- The system shall display all S019 and S036 Certification System certificates for the student: LPT Certificate, Qualifying Certificate, and course completion certificates.
- Each certificate supports view, download (PDF-A), and share, with sharing generating a time-limited S019.5 verification link.
- Multiple certificates are clearly distinguished by type and date.

Validation

- The certificate share link expires in 90 days and is renewable from SA.05.

Workflow

- Certificate issued in S019/S036 → appears in SA.05 Certificates Wallet → student views, downloads as PDF-A, or shares via a generated 90-day verification link → link renewable when needed.

Acceptance Criteria

- Given a certificate is issued, when the student opens SA.05, then it appears with view, download, and share options.
- Given a student shares a certificate, when the link is generated, then it is time-limited to 90 days and renewable.

SRS-S028-B-07: Real-Time NBE Exam Registration

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
B-07	SA.06 Exam Registration should allow eligible students to initiate NBE registration, term/year exams registration	Eligibility check is real-time	SHOULD

User Story

As an eligible student, I want to register for the Bar exam directly from the app, so that I don't need a separate process once I've met the eligibility requirements.

Requirements

- SA.06 should allow eligible students to initiate NBE registration directly from S028.
- S027 NLEMS NBE_ELIGIBLE status is checked in real time before registration is enabled.
- Registration routes to S063 NBES; the admission slip and exam centre assignment are retrievable from SA.06.

Validation

- The eligibility check must be real-time - eligibility status must never be served from a cache for this purpose.

Workflow

- Student opens SA.06 → S027 NBE_ELIGIBLE checked in real time → if eligible, registration enabled and routed to S063 NBES → admission slip and centre assignment become retrievable from SA.06.

Acceptance Criteria

- Given a student is not yet NBE_ELIGIBLE, when they open SA.06, then registration remains disabled.
- Given a student becomes NBE_ELIGIBLE, when they open SA.06, then registration is enabled based on a real-time check.

MODULE: Fees & Payments (SA.09)

FUNCTIONAL REQUIREMENT ID: S028-FR-C

SRS-S028-C-01: Fee Schedule Display and Payment Initiation

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
C-01	SA.09 must display the student's fee schedule and outstanding balances from S141 Payment System, itemised by fee type. Students initiate payment from SA.09, which routes to S141 for processing; S028 receives only a confirmed-payment callback - no card data is stored in S028.	TOTP required for payment initiation; PCI boundary sits at S141.	MUST

User Story

As a student, I want to see what I owe and pay it from my phone, so that managing my fees doesn't require a separate visit or system.

Requirements

- SA.09 displays the student's fee schedule and outstanding balances from S141 Payment System, itemised by fee type (tuition, registration, hostel, library, etc.).
- Students can initiate payment from SA.09, which routes to S141 for processing.
- S028 receives only a confirmed-payment callback - no card data is stored in S028.

Validation

- TOTP is required for payment initiation; the PCI compliance boundary sits entirely at S141, not S028.

Workflow

- Student opens SA.09 → views itemised fee schedule and balance → initiates payment → TOTP step-up confirms → payment routed to S141 → S028 receives confirmation callback only.

Acceptance Criteria

- Given a student initiates payment, when they proceed, then TOTP step-up is required before the payment routes to S141.
- Given a payment is processed, when S028 receives the callback, then no card data is stored in S028.

SRS-S028-C-02: Payment Receipts, History, and Balance Freshness

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
C-02	Fee payment receipts must be downloadable from SA.09 as PDF after confirmed payment. Payment history is viewable for all academic years. Fee schedule is viewable offline (cached). Outstanding balances update within 30 minutes of a confirmed payment.	Receipt PDF cached after download; 30-minute balance update SLA from S141 confirmation.	MUST

User Story

As a student, I want downloadable receipts and an up-to-date balance, so that I have accurate records and know my current financial standing.

Requirements

- Fee payment receipts are downloadable from SA.09 as PDF after confirmed payment.
- Payment history is viewable for all academic years.
- The fee schedule is viewable offline (cached).
- Outstanding balances updated within 30 minutes of a confirmed payment.

Validation

- The receipt PDF is cached after download; the balance update SLA is 30 minutes from the S141 payment confirmation.

Workflow

- Payment confirmed in S141 → receipt PDF generated and made downloadable in SA.09 → outstanding balance updated within 30 minutes → payment history entry added, viewable across all academic years.

Acceptance Criteria

- Given a payment is confirmed, when 30 minutes pass, then the outstanding balance reflects the payment.
- Given a student downloads a receipt, when they go offline later, then the receipt PDF remains accessible.

SRS-S028-C-03: Fee Payment Reminders

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
C-03	The shall support fee instalment plan display: if a student is on an approved S141 instalment plan.Displays the schedule, upcoming due dates, and amounts due, with reminder notifications via SA.08 7 days and 1 day before each due date. [Installment plan will not be on the payment System]	Instalment reminders are SA.08 notifications - a student welfare function.	SHOULD

User Story

As a **student**, I want to view my outstanding fee payment deadlines and receive reminders before they are due so that I can make payments on time.

Requirements

- SA.09 shall display upcoming fee payment due dates and outstanding amounts.
- SA.08 shall send reminder notifications 7 days and 1 day before each fee payment due date.

Validation

- Fee payment reminders shall be delivered through SA.08 notifications.

Workflow

- Fee payment due date is established → SA.09 displays the due date and amount payable → SA.08 sends reminders 7 days and 1 day before the due date.

Acceptance Criteria

- **Given** a student has an upcoming fee payment due date, **when** they access SA.09, then the due date and amount payable are displayed.
- **Given** a fee payment due date is approaching, when 7 days or 1 day remain, then SA.08 sends the appropriate reminder notification.

MODULE: Hostel & Library (SA.10–SA.11)

FUNCTIONAL REQUIREMENT ID: S028-FR-D

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
D-01	Student Accommodation & Library shall display the student's hostel/hall assignment (hall/hostel, block, room, room occupants subject to privacy rules), hostel/hall notices, and assigned reading list. SA.10 shall also provide a link that redirects the student to the institution's web-based Library Management System.	Hostel/hall details and reading list are displayed, and selecting Open Library redirects the student to the Library Management System.	MUST

SRS-S028-D-01: Hostel/Hall Information and Library Access

User Story

As a student, I want to view my hostel or hall details, access my reading list, and open the Library Management System so that I can easily access accommodation and library information.

Requirements

SA.10 shall display the student's current hostel or hall assignment, including:

- Hall/Hostel name
- Block
- Room number
- Room occupants (subject to institutional privacy policies)
- Hostel/Hall notices

SA.10 shall display the student's assigned reading list.

SA.10 shall provide an Open Library option that redirects the student to the institution's web-based Library Management System.

Validation

- Room occupant information shall be displayed in accordance with institutional privacy policies.

- The reading list shall be retrieved from the integrated Library Management System.
- Selecting Open Library shall redirect the student to the Library Management System.

Workflow

- Student opens SA.10 → Hostel/Hall information and reading list are displayed → Student selects Open Library → System redirects the student to the Library Management System.

Acceptance Criteria

- **Given** a student has a hostel or hall assignment, when they access SA.10, **then** their accommodation details are displayed.
- **Given** a student accesses SA.10, when the page loads, **then** their reading list is displayed.
- **Given** a student selects Open Library, when the action is performed, then the system redirects the student to the web-based Library Management System.

MODULE: Welfare & Counselling (SA.12) - 47

FUNCTIONAL REQUIREMENT ID: S028-FR-E

SRS-S028-E-01: Welfare Self-Referral, Status View, and Messaging

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
E-01	SA.12 must provide: a self-referral form (authenticated or anonymous) routed to S031 Welfare CMS; own case status view (OPEN/CLOSED, no content); secure in-app messaging with the assigned welfare officer; and a crisis button on every SA.12 screen and the app home screen. WCAG 2.1 AA on SA.12 is a statutory safeguarding requirement.	Anonymous mode requires no S001 IAM auth; crisis button reachable in ≤ 2 taps; welfare status view works fully offline.	MUST

User Story

As a student, I want to refer myself for welfare support, check my case status, and message my welfare officer, all from the app, so that getting help is as easy as using any other feature.

Requirements

- SA.12 provides a self-referral form, available in authenticated or anonymous mode, routed to S031 Welfare CMS.
- SA.12 provides an own-case status view (OPEN / CLOSED only - no case content).
- SA.12 provides secure in-app messaging with the assigned welfare officer.
- A crisis button is visible on every SA.12 screen and accessible from the app home screen, linking directly to crisis intake or emergency services.

Validation

- WCAG 2.1 AA compliance on SA.12 is a statutory safeguarding requirement, not a general accessibility goal alone.

Workflow

- Student opens SA.12 (authenticated or anonymous) → submits self-referral or checks case status → messages assigned welfare officer securely → crisis button always visible for immediate escalation.

Acceptance Criteria

- Given a student views their case status, when SA.12 loads, then only OPEN/CLOSED status is shown - no case content.
- Given a student is on any SA.12 screen, when they look for help, then a crisis button is visible.

MODULE: Scholarships (SA.13)

FUNCTIONAL REQUIREMENT ID: S028-FR-F

SRS-S028-F-01: Eligibility-Filtered Scholarship Listings and Applications

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
F-01	The system shall display all available scholarships filtered by eligibility (academic standing from S027, programme, year of study). Students can apply, upload documents, track application status,	Eligibility pre-filter shows only scholarships the student qualifies for, reducing irrelevant applications.	MUST

	and receive award/rejection notifications. (From the FMS)		
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User Story

As a student, I want to see only the scholarships I'm eligible for and apply directly, so that I don't waste time on opportunities I don't qualify for.

Requirements

- SA.13 displays all available scholarships from the GSL Scholarship Management System, filtered by the student's eligibility (academic standing from S027 NLEMS, programme, year of study).
- Students can apply, upload supporting documents, track application status, and receive scholarship award or rejection notifications via SA.08.

Validation

- The eligibility pre-filter shows only scholarships the student qualifies for based on S027 academic standing, reducing irrelevant applications.

Workflow

- Student opens SA.13 → sees only eligible scholarships → applies and uploads documents → tracks status → receives award or rejection notification via SA.08.

Acceptance Criteria

- Given a student does not meet a scholarship's academic standing requirement, when they browse SA.13, then that scholarship is not shown to them.
- Given a student applies for a scholarship, when a decision is made, then they receive a notification via SA.08.

MODULE: Notifications & Communications (SA.08)

FUNCTIONAL REQUIREMENT ID: S028-FR-G

SRS-S028-G-01: Central Categorised Notification Inbox

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
G-01	SA.08 must serve as the central inbox for all CLET-originated communications (academic, welfare status-only, fees, library, hostel, scholarship, events), categorised and filterable by type.	Push delivery requires connectivity; notifications cached in the inbox on receipt and accessible offline.	MUST

User Story

As a student, I want one place to see all my CLET notifications, so that I don't miss important updates scattered across different channels.

Requirements

- SA.08 serves as the central inbox for all CLET-originated communications: academic, welfare (status only), fees, library, hostel, scholarship, and events.
- All notifications are categorised and filterable by type.

Validation

- Push delivery requires connectivity, but notifications are cached in the SA.08 inbox on receipt and remain accessible offline.

Workflow

- Event occurs in a source system → notification generated and pushed → cached in SA.08 inbox on receipt → student filters and reviews by category, online or offline.

Acceptance Criteria

- Given a notification is received, when the student goes offline afterward, then it remains visible in the SA.08 inbox.
- Given multiple notification categories exist, when the student filters, then only the selected category is shown.

SRS-S028-G-02: Granular Notification Preferences with Statutory Exceptions

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
G-02	S028 must support granular notification preferences: students can opt out of specific categories (e.g. society events) without affecting statutory notifications (academic	Statutory notifications cannot be disabled: results, admission decisions, exam notices, welfare safety alerts.	MUST

	results, fee reminders, welfare alerts), which are defined by the GSL Student Support Administrator and cannot be opted out of.		
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User Story

As a student, I want to control which notifications I receive, while trusting that critical alerts will always reach me regardless of my preferences.

Requirements

- S028 supports granular notification preferences: students can opt out of specific categories (e.g. society events).
- Statutory notifications - defined by the GSL Student Support Administrator - cannot be opted out of.

Validation

- Statutory notifications include results, admission decisions, examination notices, and welfare safety alerts - none of these are opt-outable under any student preference setting.

Workflow

- Student adjusts notification preferences → non-statutory categories can be toggled off → statutory categories remain enforced regardless of preference settings.

Acceptance Criteria

- Given a student attempts to disable a statutory notification category, when they try, then the option is not available.
- Given a student opts out of society event notifications, when a society event notification is triggered, then it is not delivered to that student.

SRS-S028-G-03: Timely, Content-Minimised Push Delivery

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
G-03	Push notifications must be delivered within 5 minutes of the triggering event. Welfare-related push notifications must not include case details - only 'You have a welfare update' with a deep link to SA.12.	Welfare notification content minimisation is both a Ghana DPA 2012 and a confidentiality requirement.	MUST

User Story

As a student, I want to be notified quickly, but I don't want sensitive details about my welfare case appearing on my lock screen.

Requirements

- Push notifications for SA.08 are delivered within 5 minutes of the triggering event in the source system.
- Welfare-related push notifications must not include case details - only 'You have a welfare update' with a deep link to SA.12.

Validation

- Welfare notification content minimisation is required both under Ghana DPA 2012 and as a confidentiality obligation - no case type, officer name, or status may appear in the push payload.

Workflow

- Triggering event occurs in source system → push notification generated within 5 minutes → for welfare events, payload is minimised to a generic message and deep link only.

Acceptance Criteria

- Given a triggering event occurs, when the notification is generated, then it is delivered within 5 minutes.
- Given a welfare-related notification is sent, when its payload is inspected, then it contains no case details.

SRS-S028-G-04: Opt-In Weekly Digest Notifications

REQ-ID	REQUIREMENT	ACCEPTANCE CRITERIA	PRIORITY
G-04	S028 should support scheduled digest notifications: a weekly summary showing outstanding items across all SA modules (unpaid fees, deadlines, overdue library items, pending appeals). Digest is opt-in and does not disable individual notifications.	Weekly digest reduces notification fatigue while ensuring students do not miss important items.	SHOULD

User Story

As a student who doesn't want frequent notifications, I want a single weekly summary of what needs my attention, so that I stay on top of things without notification fatigue.

Requirements

- S028 should support scheduled digest notifications: a weekly summary showing outstanding items across all SA modules - unpaid fees, approaching assignment deadlines, overdue library items, pending appeals.
- The digest is opt-in.
- Individual notifications are not disabled by digest opt-in.

Validation

- Opting into the digest must not reduce or replace any individual notification the student would otherwise receive.

Workflow

- Student opts into the weekly digest → S028 compiles outstanding items across all modules → digest notification delivered weekly → individual notifications continue to be delivered separately.

Acceptance Criteria

- Given a student opts into the digest, when the week ends, then a summary notification of outstanding items is delivered.
- Given a student has the digest enabled, when an individual notification event occurs, then it is still delivered separately.

4. Data Requirements

4.1 Key Data Entities

- Student Profile (S028 Core, tokenised): CLET_ID, name, programme, year of study, notification preferences, roommate-consent flag. Resolved to backend records via S001 IAM and S026 - not held as a duplicate academic record.
- Admissions Status (SA.01, from S027): status (Application Received / Under Review / Decision Pending / Offer Made / Offer Accepted / Enrolled), case reference, decision detail.
- Timetable Cache (SA.02, from S029): session name, lecturer, room/venue, date/time, session type, cached for 7-day offline access.
- Attendance Record (SA.03, to S029): check-in method (QR/geo), timestamp, session reference; queued offline entries are idempotently synced.

- Academic Record Cache (SA.04, from S027): module, assessment type, mark, grade, cumulative GPA, and the downloaded transcript PDF.
- Certificate Record (SA.05, from S019/S036): certificate type, issue date, PDF-A file, and the time-limited (90-day) verification share link.
- Fee Record (SA.09, from S141): itemised fee schedule, outstanding balance, payment history, receipts, and instalment schedule where applicable. No card data is stored in S028.
- Hostel Record (SA.10): block, room, roommate name (contact details only if consented), maintenance request history.
- Library Record (SA.11, from S104): borrowed items, due dates, reservation status, fine status, and the offline-accessible library card barcode.
- Welfare Case Status (SA.12, from S031): OPEN/CLOSED status only - no case content - plus encrypted message threads readable only on the student's and welfare officer's own devices.
- Scholarship Application (SA.13): eligibility-filtered listing, application status, award status. Supporting documents are transmitted over TLS and not retained in S028 beyond the upload session.
- Notification Inbox (SA.08, from S025 / Push Notification Service): category, status, deep link, read/unread flag - cached locally, with sensitive content excluded from the payload.

4.2 Local Cache and Retention Constraints

- Only data required for the offline capabilities defined in the App Modules Reference may be cached locally; no welfare case content and no financial details beyond balance and receipt summary may be stored on-device.
- All local cache data is stored using platform encrypted storage APIs (iOS Data Protection / Android Keystore); no cached academic, welfare, or financial data may be stored in plaintext.
- Biometric data never enters S028's data model at all - it remains solely within the device's Secure Enclave, managed by platform APIs.

5. External Interface Requirements

5.1 Integration Systems

- S026 API Gateway - the single integration point for all S028 backend calls. No direct API calls from S028 to backend databases are permitted. S026 enforces schema validation, field-level access control, rate limiting, certificate pinning, and S003 audit logging for every S028 integration.

- S027 NLEMS - admissions status (27.02, 27.03), cohort assignment, results and transcript (27.10, 27.12), progression (27.05), NBE eligibility (27.11), and academic standing for scholarship pre-filtering. S028 requests only the fields required for the SA module being loaded - never a full student record on app launch.
- S029 LMS - timetable feed (SA.02) and the attendance module (16.9, SA.03), with attendance check-ins synced from S028's offline queue.
- S141 Payment System - fee schedule, payment processing, and receipts (SA.09). S028 stores no card data; the PCI compliance boundary sits entirely at S141.
- S104 Library System - borrowed items, due dates, reservations, and fines (SA.11).
- S031 Welfare CMS - welfare self-referral (authenticated or anonymous), case status (OPEN/CLOSED only), and end-to-end encrypted secure messaging (SA.12). S026 enforces that no case content crosses into S028's API responses.
- S019 / S036 Certification Systems - certificate issuance and the time-limited S019.5 verification link (SA.05).
- S063 NBES - Bar exam registration, admission slip, and exam centre assignment (SA.06), gated on a real-time S027 NBE_ELIGIBLE check.
- S009 Tribunal System - onward tribunal referral submissions from SA.07 Appeals.
- GSL Scholarship Management System - scholarship listings, applications, document uploads, and appeals (SA.13).
- GSL Hostel Management System (S120 or equivalent) - hostel assignment, rules, and maintenance requests (SA.10).
- S043 Society CMS - society membership, events, and the S043 Voting Engine for anonymous ballots (SA.14).
- S025 Communications System and Push Notification Service - all SA.08 notification delivery, with content-minimised payloads for welfare, grades, and fees.
- S001 IAM - biometric/PIN authentication, SSO across all integrated backend systems, TOTP step-up for sensitive actions, and CLET DG step-up MFA for major release approval.
- S003 Audit - logging of Admin Portal configuration changes and app release events; no student personal data is logged.
- App Store / Play Store - the distribution channels for S028 releases, each requiring a WCAG 2.1 AA audit report before submission.

6. Non-Functional Requirements

6.1 Accessibility (Statutory under Act 47)

- NFR-ACC1: All 14 SA modules must meet WCAG 2.1 Level AA on both iOS and Android - a statutory requirement, not a best-practice guideline. No waiver, phased compliance, or 'best effort' interpretation is permitted; non-compliant screens must be fixed before release.
- NFR-ACC2: Full iOS VoiceOver and Android TalkBack support on all screens - descriptive labels on all interactive elements, alt text on all images, accessible form field labels, and full keyboard navigability, tested on both platforms before each release.
- NFR-ACC3: Dynamic text size support - all screens must reflow legibly at maximum device accessibility text size (200% iOS, 130sp Android) without horizontal scrolling or truncation; fixed-size font implementations are prohibited.
- NFR-ACC4: High-contrast mode support on both platforms, with a minimum 4.5:1 contrast ratio in normal mode and 7:1 in high-contrast mode; colour must never be the sole means of conveying information.
- NFR-ACC5: SA.12 Welfare must meet an enhanced standard: the crisis button reachable in ≤ 2 taps from any screen, the crisis form fully completable with VoiceOver/TalkBack active, and emergency contact numbers displayed in large text without requiring zoom.

6.2 Performance

- NFR-P1: All primary S028 screens load within 1 second on a 3G connection at p95; timetable, results, and the notification inbox load within 500ms on 3G when displaying cached data.
- NFR-P2: A stale-data indicator is displayed whenever cached data exceeds its screen's freshness threshold (timetable 7 days, results 24 hours, fee balances 30 minutes, library due dates 24 hours); stale data must never be presented as current without the indicator.

6.3 Security

- NFR-S1: Biometric data is stored exclusively in the device's Secure Enclave and never transmitted over any network or stored on any CLET server; only platform biometric APIs (Apple LocalAuthentication, Android BiometricPrompt) may be used - no custom implementations.
- NFR-S2: All backend communication uses TLS 1.3 minimum, with certificate pinning on all S026 API Gateway endpoints; HTTP is prohibited on all S028 network calls.
- NFR-S3: SA.09 Fees & Payments and SA.12 Welfare messages require TOTP re-authentication for write actions; SA.09 enforces a 60-second payment confirmation timeout, after which the payment session expires and must be restarted.

6.4 Privacy

- NFR-PR1: Only data required for the offline capabilities in the App Modules Reference may be cached locally; no welfare case content and no financial details beyond balance and receipt summary may be stored on the device.
- NFR-PR2: Push notification payloads contain no sensitive personal data - welfare notifications read only 'You have a welfare update', grade notifications read only 'Your results are available', and fee notifications may show an outstanding amount but never an itemised breakdown.

6.5 Availability

- NFR-AV1: Offline-capable features (timetable, grades, cached notifications, library barcode, welfare crisis contacts) remain functional even when all backend systems are unavailable; the app must never show a blank screen - always cached data or a clear offline message with a last-updated timestamp.
- NFR-AV2: The S026 API Gateway must achieve 99.9% uptime; app store deployments maintain an 18-month backward compatibility window so students on older app versions retain access to core academic services.

6.6 Offline / Sync

- NFR-OFF1: S028 uses a local-first data architecture - reads served from local cache with freshness indicators; writes queued offline and synced on reconnect. Queued writes (attendance check-ins, welfare referrals, fee payments) must be idempotent, never duplicating on sync.
- NFR-OFF2: Offline sync is transparent to the student - a sync progress indicator is shown, SA.08 confirms each queued action once the backend acknowledges it, and failed syncs produce a clear error notification rather than silently losing student-submitted data.

6.7 Usability

- NFR-U1: The home screen displays a personalised dashboard - today's timetable, outstanding actions (unpaid fees, deadlines, library returns due), and unread notification count - all three visible without scrolling on a standard 6-inch smartphone display.
- NFR-U2: S028 supports left-to-right and right-to-left text layout via the platform's native layout direction APIs, not fixed-position pixel layouts, to support future multilingual additions.

6.8 Notification

- NFR-NOTIF1: Push notifications are delivered within 5 minutes generally, within 2 minutes for exam-related notifications, and immediately (0-delay, highest priority) for welfare safety alerts.
- NFR-NOTIF2: Device Do Not Disturb settings are respected for non-urgent notifications; welfare safety alerts, examination notices, and fee overdue notices may override DND only if the student has explicitly enabled critical alerts in S028 settings - never enabled by default.

6.9 Constraints and Dependencies

- Legal / 47 (CON-L1–L3): WCAG 2.1 AA is a statutory requirement for every screen, flow, notification, and document S028 produces - a non-compliant release is unlawful; the pre-release accessibility audit must be conducted by an independent assessor on physical devices and signed off by the CLET DG; and the SA.12 anonymous crisis pathway must never be login-gated and cannot be removed in any future release. Owner: CLET DG + DTI Technical Lead / GSL Student Support Administrator.
- Accessibility (CON-ACC1–ACC2): every screen must be fully navigable by keyboard and switch access, with no interaction reachable only by gesture; and timed interactions (e.g. the 60-second payment timeout) must provide a warning, an extension option, and the ability to turn the timeout off, per WCAG 2.1 AA Success Criterion 2.2.1. Owner: DTI Technical Lead.
- Privacy / DPA (CON-PR1–PR2): biometric data must never leave the student's device - no CLET server or API layer may receive it; and push notification payloads must contain no sensitive personal data, since notifications are briefly visible on the lock screen. Owner: DTI Technical Lead + CLET DPO.
- Technical (CON-T1–T3): S026 API Gateway is the sole integration point with all backend systems - no direct database calls are permitted; local cache must use platform encrypted storage APIs with no plaintext sensitive data at rest; and an 18-month backward compatibility window must be maintained for every S026 endpoint used by S028. Owner: DTI Technical Lead + Platform Architecture.
- Integration (CON-I1–I2): S028 reads from S027 must respect data minimisation - only the fields required for the specific SA module being loaded, never a full record on launch; and the S031 welfare integration must expose only OPEN/CLOSED case status, enforced at the S026 response schema level. Owner: DTI Technical Lead + Platform Architecture / CLET DPO.
- Governance (CON-G1): S028 is co-owned by GSL Student Support Services (product decisions affecting statutory obligations) and DTI (technical delivery decisions); major releases require CLET DG step-up MFA approval. Owner: GSL Student Support + DTI joint governance.
- Phase (CON-P1): as a FAST-TRACK delivery, S028's backend integrations must not be blocked by Phase 2/3 system readiness - if S029 LMS is unavailable at launch, SA.02 falls back to manual timetable import; if S031 Welfare CMS is unavailable, SA.12 operates in crisis-only mode (anonymous referral only). Owner: DTI Technical Lead + Programme Architecture.

7. Appendix A: Glossary

Glossary of Terms

- SA - the module code prefix for the 14 Student App screens (SA.01 Admissions Status through SA.14 Co-curricular / Society).
- 47 - Legal Education Act 47, the statutory basis mandating WCAG 2.1 AA accessibility for S028.
- WCAG 2.1 AA - Web Content Accessibility Guidelines, Level AA; a statutory compliance target for every S028 screen, flow, and document.
- DTI - Digital Transformation Initiative, S028's technical delivery and infrastructure co-owner.
- Secure Enclave - the on-device hardware security module storing biometric data; never transmitted off the device.
- SSO - Single Sign-On via S001 IAM, giving authenticated access to all integrated backend systems without re-authentication.
- TOTP - Time-based One-Time Password, required for sensitive actions such as fee payment, appeal submission, and welfare self-referral.
- NBE - National Bar Examination, administered via S063 NBES and gated on S027 NBE_ELIGIBLE status.
- S026 API Gateway - the single, schema-enforcing integration point between S028 and all backend systems.
- Crisis button - the persistent SA.12 shortcut reachable in ≤ 2 taps from any screen, usable without login, connecting directly to welfare crisis intake or emergency services.
- Local-first architecture - the offline-first data design where reads are served from cache and writes are queued offline and synced idempotently on reconnect.
- Feature flag - a mechanism to deploy a new feature to a subset of students before full rollout, allowing rapid rollback if accessibility or stability issues arise.
- Stale-data indicator - the on-screen marker shown whenever cached data exceeds its defined freshness threshold.
- S001 IAM / S003 Audit / S025 Communications System - central supporting enterprise services used across S028.

